DQN extensions in a paired randomised trial on MinAtar

Overview of current results — draft

Draft, 16 October 2026. PER training-loss histories are being regenerated.

Summary

This draft reports the corrected trial (`corrected-rct-01`): four extensions of DQN, each compared with standard DQN on the same 30 sampled hyperparameter configurations, in four MinAtar games (600 completed runs). The primary outcome is the area under the evaluation learning curve (AUC).

- **Averaged DQN** has a significantly higher AUC than DQN in Asterix (+57.5%), Freeway (+20.9%) and Space Invaders (+19.5%). It does better on 24–26 of the 30 configurations in each of these games.
- **Double DQN** is significantly better only in Space Invaders (+11.4%). Its other differences are not significant.
- **PER** is significantly worse in Freeway (−35.3%) and Space Invaders (−50.0%). Fixing the priority-bookkeeping bug barely changed PER (see *Did the PER fix change PER?*), so the bug did not cause the deficit.
- **n-step DQN** is significantly worse in Asterix (−80.0%), Freeway (−59.4%) and Space Invaders (−21.0%). In Asterix it hardly learns (Figure 1).
- **Breakout** shows no significant difference for any extension. All four 98.75% intervals include zero. This does not show that the extensions are equivalent there.
- All nine significant results under the planned per-game threshold remain significant with Bonferroni correction across all 16 comparisons, and the Wilcoxon signed-rank test agrees with each of them.

Design

Arms. DQN (control), Double DQN, prioritised experience replay (PER; proportional, $\alpha = 0.6$, $\beta_0 = 0.4$), n-step DQN ($n = 3$) and Averaged DQN ($K = 10$).

Individuals. 30 individuals were drawn once from the shared hyperparameter space (learning rate, buffer size, batch size, target-update interval, ε-decay length, network depth). Each has its own training seed. Every arm trains the same 30 configuration–seed pairs, so the comparisons are paired. The statistical unit is the individual. The ten evaluation episodes and the 100 checkpoints are repeated measurements, not replicates.

Protocol. Each run trains for 5 million frames. Every 50,000 frames the greedy policy is evaluated over ten episodes with fixed seeds. AUC is the trapezoidal integral of mean evaluation return from 50,000 to 5,000,000 frames, reported in 10^6 return $\times$ frames.

Inference. Each treatment is compared with DQN using a two-sided paired t-test on the 30 AUC differences. The proposal sets a Bonferroni threshold of $\alpha = 0.05/4 = 0.0125$ within each game, with matching 98.75% confidence intervals.

Data and provenance

- Double DQN, PER, n-step and Averaged DQN results come from `Results/corrected-rct-01`.
- The corrected run did not include DQN, so the DQN results come from the original run (`Results/original-run-2026-09`). This is valid because the corrected run reproduced every Double DQN, n-step and Averaged DQN trajectory bit for bit. Between the two runs, the only code change that affects learning is the PER fix.
- n-step Freeway individuals 18–29 also come from the original run, because the corrected job failed at start-up. Individuals 0–17 match in both runs. Restricted to individuals 0–17, the effect is -53.3% ($p = 2.4 \times 10^{-5}$), which supports the full-sample result.
- Every stored AUC recomputes exactly from its trajectory. Each individual has the same seed and configuration in every arm. Independent recomputation reproduces all 16 paired tests.
- Seaquest is excluded. In every arm, a greedy evaluation episode stalled beyond the 100,000-step guard, which aborts the run on purpose.

Primary outcome

Table 1: Mean AUC (10^6 return $\times$ frames, $n = 30$ per cell). Parentheses show the change relative to DQN, computed from unrounded means. Bold marks the highest observed mean in each game. It does not indicate significance.

Arm	Breakout	Asterix	Freeway	Space Invaders
DQN (control)	65.21	16.77	159.81	168.84
Double DQN	65.31 (+0.1%)	21.49 (+28.2%)	146.18 (−8.5%)	188.03 (+11.4%)
PER	66.69 (+2.3%)	13.99 (−16.5%)	103.36 (−35.3%)	84.39 (−50.0%)
n-step DQN	**68.13 (+4.5%)**	3.35 (−80.0%)	64.86 (−59.4%)	133.38 (−21.0%)
Averaged DQN	65.91 (+1.1%)	**26.41 (+57.5%)**	**193.21 (+20.9%)**	**201.72 (+19.5%)**

Table 2 and Figure 2 show the comparisons. Under the per-game threshold, nine of the 16 comparisons are significant: Averaged DQN in three games, Double DQN in one, PER in two and n-step DQN in three. Two comparisons are close but not significant: Double DQN in Asterix (+28.2%, $p = 0.048$) and Double DQN in Freeway (-8.5%, $p = 0.014$). Their intervals include zero.

In absolute terms, the differences between arms are largest in Freeway and Space Invaders. Breakout curves overlap almost entirely (Figure 1). The variation between individuals is large. In Asterix and Space Invaders, the spread of AUC within an arm is comparable to the differences between arms (Figure 3). This is why the paired design matters.

Robustness of the main results

- **Multiplicity.** The proposal fixes the family at four comparisons per game. If the family is all 16 comparisons ($\alpha = 0.003125$), all nine findings still hold. The closest is Double DQN in Space Invaders ($p = 0.0029$).

Table 2: Paired comparisons with DQN on AUC. Δ is the mean paired difference as a percentage of the DQN mean. The interval is the 98.75% t interval on the same scale. It is symmetric, so it can extend below -100%. d_z is the mean difference divided by the SD of the differences. "Better" counts individuals whose treatment AUC exceeds their DQN AUC. Bold p-values are below 0.0125.

Game	Arm	Δ (%)	98.75% CI (%)	d_z	t-test p	Wilcoxon p	Better
Breakout	Double DQN	+0.1	$[-5.1, +5.4]$	+0.01	0.943	0.919	14/30
	PER	+2.3	$[-8.5, +13.0]$	+0.10	0.579	0.360	17/30
	n-step DQN	+4.5	$[-6.1, +15.1]$	+0.20	0.271	0.245	19/30
	Averaged DQN	+1.1	$[-6.9, +9.1]$	+0.07	0.724	0.715	17/30
Asterix	Double DQN	+28.2	$[-8.2, +64.6]$	+0.38	0.048	0.129	17/30
	PER	−16.5	$[-42.4, +9.4]$	−0.31	0.100	0.271	14/30
	n-step DQN	−80.0	$[-122.4, -37.7]$	−0.92	$\mathbf{2.3 \times 10^{-5}}$	1.2×10^{-6}	2/30
	Averaged DQN	+57.5	$[+13.2, +101.8]$	+0.63	**0.0017**	0.0004	25/30
Freeway	Double DQN	−8.5	$[-17.2, +0.1]$	−0.48	0.014	0.019	10/30
	PER	−35.3	$[-52.0, -18.7]$	−1.03	$\mathbf{4.1 \times 10^{-6}}$	1.0×10^{-7}	3/30
	n-step DQN	−59.4	$[-81.5, -37.4]$	−1.31	$\mathbf{6.7 \times 10^{-8}}$	1.9×10^{-9}	0/30
	Averaged DQN	+20.9	$[+10.6, +31.1]$	+0.99	$\mathbf{7.7 \times 10^{-6}}$	1.1×10^{-5}	24/30
Space Invaders	Double DQN	+11.4	$[+2.1, +20.7]$	+0.59	**0.0029**	0.012	21/30
	PER	−50.0	$[-64.8, -35.2]$	−1.64	$\mathbf{7.1 \times 10^{-10}}$	9.3×10^{-9}	1/30
	n-step DQN	−21.0	$[-35.2, -6.8]$	−0.72	**0.0005**	0.0002	6/30
	Averaged DQN	+19.5	$[+10.3, +28.6]$	+1.03	$\mathbf{4.1 \times 10^{-6}}$	3.8×10^{-6}	26/30

- **Normality.** Shapiro–Wilk flags the paired differences ($p < 0.05$) for n-step in Breakout, Asterix and Freeway, PER in Asterix and Freeway, and Double DQN in Space Invaders. Each flagged significant result is matched by the Wilcoxon signed-rank test. The flagged non-significant comparisons stay non-significant under it.
- **Other endpoints.** With mean return over the last ten evaluations, eight of the nine findings remain significant, all in the same direction. The exception is n-step in Space Invaders (-12.1%, $p = 0.048$). Double DQN in Asterix also becomes significant ($+53.9\%$, $p = 0.007$). With return at the final checkpoint alone, which is a single noisy evaluation, four of the nine remain significant. These are exploratory secondary endpoints.

Did the PER fix change PER?

The original run used a PER implementation with a bug in how maximum priorities were tracked. The corrected run retrained PER on the same 30 individuals with the bug fixed and left all other code unchanged.

Table 3: PER mean AUC (10^6) before and after the fix, with the paired t-test between the two runs on the same individuals.

Game	Before fix	After fix	Change	Paired p
Breakout	65.29	66.69	+2.1%	0.58
Asterix	14.12	13.99	−0.9%	0.89
Freeway	114.11	103.36	−9.4%	0.27
Space Invaders	80.97	84.39	+4.2%	0.43

None of the changes is significant (Table 3), and the conclusions about PER versus DQN are the same before and after the fix. So the PER deficit in Freeway and Space Invaders comes from something other than the bug. Candidates include the interaction with the shared training recipe (MSE loss, RMSprop) and the importance weights. The Space Invaders curve drops in the first few hundred thousand frames

and stays at about half of DQN's level (Figure 1). The loss histories being regenerated are needed to examine this.

Limitations and open items

- **PER loss histories (in progress).** Individuals 0–12 of PER have results but no loss log. They are being retrained with identical code, environment and seeds. A rerun is accepted only if its evaluation curve matches the reported one exactly. PER has already reproduced exactly across separate cluster jobs.
- **Loss coverage elsewhere.** DQN loss histories exist only in the original run's logs. Commit `d697076` removed them from the working tree, but git history still has them. In the corrected run, Double DQN Breakout has loss histories for only 2 of 30 individuals. Loss figures must state which individuals they cover.
- **Seaquest** is excluded (see Data and provenance). Reporting it would need a protocol change, such as a capped evaluation that records partial scores, applied to every arm.
- **Scope.** Each configuration has one training seed, so sensitivity to the configuration cannot be separated from seed variation. The training recipe is shared across arms and is not each paper's published recipe. The results concern these variants under this recipe on MinAtar.
- **AUC** covers only the observed interval and mixes learning speed with final level. The learning curves and the secondary endpoints should be read alongside it.

Next steps

1. Finish the PER loss rerun, check it, and add PER loss diagnostics: the weighted loss, unweighted TD error, and importance-weight statistics, which are already logged.
2. Run exploratory analyses that need no new training. These include whether the treatment effects depend on the hyperparameters (e.g. n-step against buffer size and learning rate), ranking stability across AUC, the last-ten-evaluation mean and the best checkpoint, and paired-bootstrap intervals.
3. Decide on the inference family (per game or across games) and on how Seaquest is reported, then freeze the analysis plan before writing the results chapter.

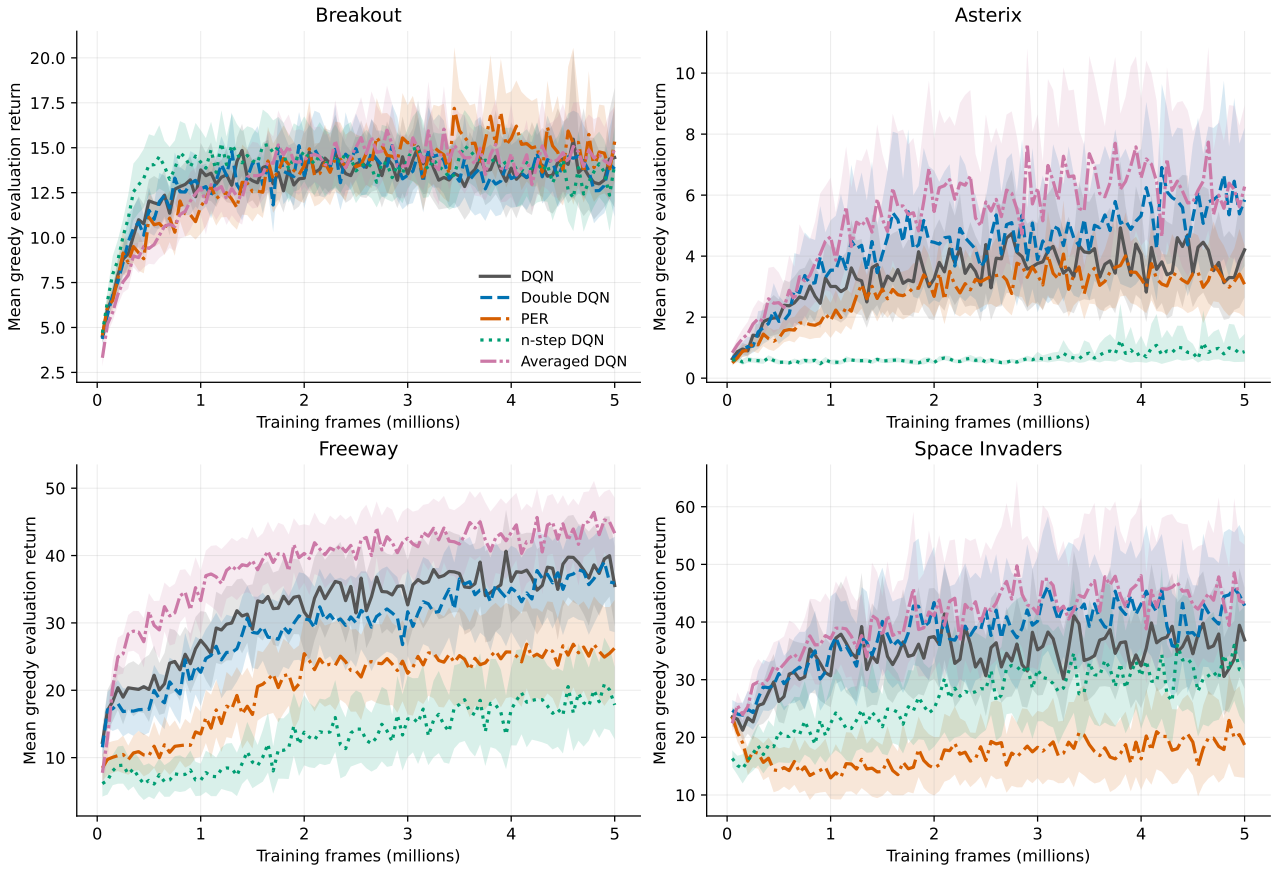


Figure 1: Mean greedy evaluation return during training ($n = 30$ per arm). Shading shows pointwise 95% bootstrap intervals from resampling whole individuals. These are not simultaneous bands. Curves are not smoothed.

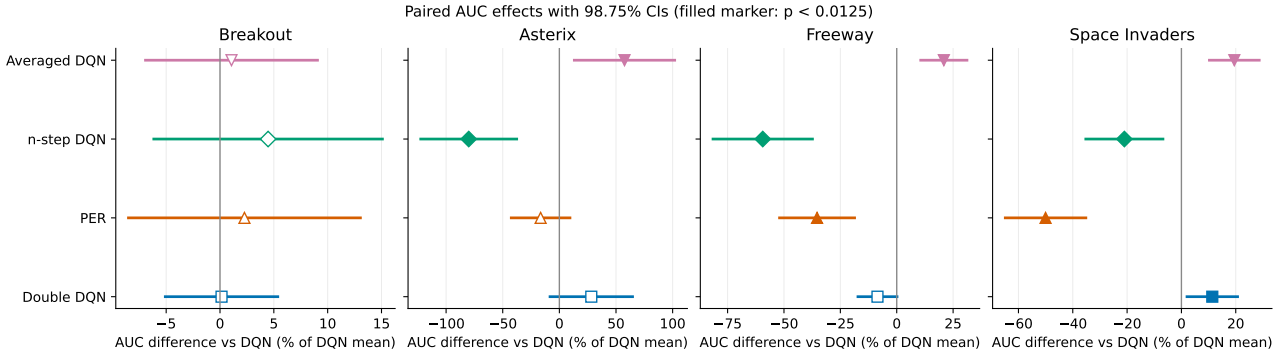


Figure 2: Mean paired AUC difference from DQN, as a percentage of the DQN mean, with 98.75% t intervals. Filled markers have $p < 0.0125$. Positive values favour the treatment.

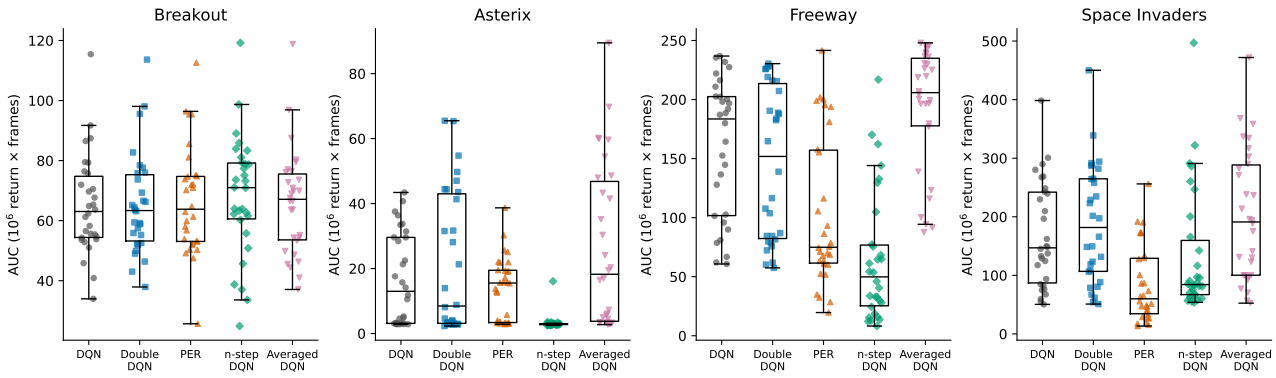


Figure 3: AUC of every individual by arm. Boxes show medians and interquartile ranges. All 30 runs per arm are plotted.